Replication of the orthogonal latent-mean contrasts in Jeffries (2024)

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Abstract

Do the latent-mean contrasts in Jeffries (2024) sit on the printed scale −0.40, −0.20, 0, and do the rejection rates and the MIMIC convergence counts follow from that model? Jeffries defines three codes for four groups and reports multigroup and MIMIC averages near those three numbers when the generating means are 1, 0.6, 0.8, and 0.4, from 10,000 converged replicates in each cell. Fitting the marker model to the population that lavaan [2] builds from the generation string returns exactly −0.40, −0.20, and 0. The unbalanced power rates at $N = 240$ come out 49.35, 21.28, and 8.51 against the printed 49.23, 21.08, and 7.77. Of thirty rejection-rate comparisons, one stays outside an uncorrected two-standard-error band and none miss Bonferroni or Holm. That rate is the MIMIC model, shares 45/45/5/5, null means, contrast 2: 6.51 on the simsem 0.5.16 [3] stream and 6.47 on the hand-written stream, against the printed 7.41. The combined standard error is 0.36 percentage points, and the two-standard-error band around 7.41 runs from 6.69 to 8.13. The MIMIC convergence gap closes with lavaan's BFGS optimizer. Seven of the eight cells reach the printed 10,000 of 10,000, and the simsem null cell with those shares reaches 9,999. The named optimizer, `nlminb`, converges 9,951 to 9,988 of the same replicates, so the printed count is consistent with a different optimizer or setting.

1 The question, and the headline

This section answers whether the printed contrasts, the printed rejection rates, and the MIMIC convergence counts come from the model in the dissertation. The contrasts do. The rates do, except one. The MIMIC convergence counts close when lavaan uses BFGS.

Jeffries (2024), https://digitalcommons.unl.edu/dissunl/216/, studies one factor, four indicators, and four groups. The true-difference means are 1, 0.6, 0.8, and 0.4. The null means are all 1. Sample sizes are 240, 480, and 960, with group shares 25/25/25/25, 35/35/15/15, and 45/45/5/5. The fitting program named in the text is lavaan 0.6.17. The public analysis scripts are at https://osf.io/q9yu8/.

The population fit of the marker model on the parsed generation string is −0.40, −0.20, 0. The same marker model on a population in which every loading is 0.70 is −0.28, −0.14, 0. The printed tables match the first of those. At $N = 240$ and 10,000 replicates, the unbalanced power rates match the page. One rejection rate remains outside an uncorrected two-standard-error band: the MIMIC model, shares 45/45/5/5, null means, contrast 2, 6.51 on the simsem stream and 6.47 on the hand-written stream, against the printed 7.41. A Bonferroni or Holm correction of the thirty rate comparisons leaves no miss. The MIMIC convergence gap closes under BFGS. Seven of the eight cells reach the printed 10,000 of 10,000, and the simsem null cell with shares 45/45/5/5 reaches 9,999. The optimizer the dissertation names is `nlminb`. On these replicates nlminb converges 9,951

to 9,988. The printed count of 10,000 is what a different optimizer produces, and it is consistent with an optimizer or a setting other than that named run.

The original work is the design, the codes, and the printed tables. The calculations below are ours: the moment fits, the Monte Carlo, and the code identities. The dissertation's percent columns are one Wald test per dataset at $\alpha = 0.05$. That is the design of those columns. Multiplicity is applied only to our check that a printed rate and our rate agree.

2 What we replicated

This section answers which printed claims agree with a direct calculation. Table 1 is that comparison. "Match" means the same number, the same population target, or a Monte Carlo gap no larger than two combined standard errors. "Partial" means one piece holds and another does not. "No match" means a different scale, or a rate that remains outside that two-standard-error band. "Close" means a gap sitting on the edge of that band. The combined standard error uses the printed count of 10,000 and our count of converged replicates.

Claim	Published	Ours	Verdict
Defined contrasts on $(1, 3/5, 4/5, 2/5)$	$-0.40,\ -0.20,\ 0$	$-2/5,\ -1/5,\ 0$	match
Unweighted products of the Table 2.2 codes	Equation (2.4) is zero	Proved zero	match
Imbalance makes the set nonorthogonal	The set becomes nonorthogonal	Only code 1 with code 3 fails	partial
Marker model on the parsed generation population	Averages near $-0.40,\ -0.20,\ 0$	Population fit $-0.40,\ -0.20,\ 0$	match
Every loading 0.70, then a marker loading of 1	The same averages	Population fit $-0.28,\ -0.14,\ 0$	no match
Power, Table 4.1, $N = 240$, shares 45/45/5/5	49.23, 21.08, 7.77	49.35, 21.28, 8.51	match
Null rates, Table 4.3, same cell	8.13, 8.59, 7.73	8.10, 8.33, 8.22	match
Thirty rejection rates, uncorrected	Each percent is one test at $\alpha = 0.05$	1 of 30 outside two combined SEs	partial
Same thirty rates, Bonferroni and Holm	Those tests are a single-test design	0 of 30 outside the band	match
MIMIC null, shares 45/45/5/5, contrast 2	7.41	6.51 simsem; 6.47 hand	no match
Multigroup convergence, Table 4.5, that cell	89.29%	89.82% of 10,000	match
MIMIC convergence, eight cells, BFGS	10,000 of 10,000	10,000 in seven; 9,999 in one	match

Table 1: Published claims against this replication. Monte Carlo rows use 10,000 replicates at $N = 240$. The population rows use moment matrices.

The contrast row is equations (3.3)–(3.5) on the printed means, proved in Lean [4] at `123`. The product row is equation (2.4), proved at `103`. The imbalance row is partial because sizes (a, a, b, b) leave two of the three weighted products at zero and set the other to $2(b - a)$, proved at

68. At $N = 240$ that product is -96 in condition 2 and -192 in condition 3. The third contrast is zero while the first two means differ, proved at 127. The treatment-coefficient form of the same contrasts is proved at 159. The marker identity at 131, which multiplies the contrasts by 7/10, is the every-loading-0.70 reading in the table. It is the population that misses -0.40.

The power row is the multigroup model in the smallest groups. The largest $|z|$ among those three rates is 1.85. The averages in that cell are -0.403, -0.215, and 0.0050 against -0.404, -0.218, and 0.0054. The thirty-rate rows use one normal test per rate. The uncorrected cutoff is $|z| > 1.96$, and the two-standard-error band used in the first screen is the same rule with 2 in place of 1.96. Bonferroni on thirty tests uses $z = \Phi^{-1}(1 - 0.05/30) = 2.94$, and the two-sided threshold is $z = \Phi^{-1}(1 - 0.05/60) = 3.14$. Holm sorts the thirty two-sided p-values and stops at the first failure. At 10,000 replicates the smaller two-sided p-value for the remaining rate is 0.0088, above $0.05/30 = 0.00167$, so Holm stops immediately. The other twenty-nine rejection-rate comparisons match.

The no-match row is that one rate. The combined standard error is 0.36 percentage points on both streams, using the printed count of 10,000 and our converged counts of 9,999 (simsem) and 10,000 (hand). The two-standard-error band around 7.41 runs from 6.69 to 8.13. The simsem rate 6.51 has $|z| = 2.50$. The hand-written rate 6.47 has $|z| = 2.62$. Both sit below 6.69. The convergence row is the BFGS refit in Section 5. Under nlminb the same eight cells converge in 9,951 to 9,988 replicates. Seven BFGS cells meet the printed 10,000. The eighth, the simsem null cell with shares 45/45/5/5, is 9,999. The printed count agrees with that BFGS result. It is consistent with an optimizer or a setting other than the nlminb run the text names.

A binomial test of one miss in thirty trials at $p = 0.05$ has upper tail 0.79. That figure treats the thirty indicators as independent. The three contrasts in a cell share replicates, so the count is dependent inside each cell and the binomial probability is the independent-trials calculation. The expected count under independence is 1.5. One miss is an ordinary count for that band.

3 What gave us trouble, and how it was resolved

This section answers why an earlier replay sat on -0.28 and why a short Monte Carlo looked as if many rates had missed.

3.1 The scale -0.28 against -0.40

The generation line is `eta =~ NA*i1 + 0.7*i1 + 0.7*i2 + 0.7*i3 + 0.7*i4`. In lavaan 0.6.17 that line keeps a free loading on the first indicator and drops the duplicate 0.7. `simulateData` fills the free loading with 1. The other three loadings stay at 0.70. The factor variance in the syntax is 0.50 and the residual variances are 0.51. A large sample from that string has mean about 2.00 on the first indicator when the factor mean is 1, and means about 1.70 on the other three, which is $1 + 0.70 \times 1$.

The marker model fixes the first loading at 1. On this population the factor stays on the generating scale, and the three contrasts are -0.40, -0.20, and 0. If every loading is generated at 0.70 and the first is then fixed at 1, the same contrasts are rescaled by 0.70, which is -0.28, -0.14, and 0. The printed tables match the parsed string. The moment fits are `simulation/candidate_readings.R`, and the output is `simulation/out/candidate_readings.csv`.

The page writes `i1 ~ 1*`. That token does not parse in lavaan 0.6.17 or in simsem 0.5.17. The OSF generation script, and the null model on the same page, write `i1 ~ 1*1`. The Monte Carlo uses `1*1`.

Latent variables were identified using the marker method approach, where the first indicator's loading was fixed to one ($\lambda_1 = 1$). Per recommendations from Hancock (1997), over both SEM forms, intercepts were modeled such that $\tau_{1_1} = \tau_{2_1} = \ldots \tau_{x_i}$. Further, the variance of all latent variables were freely estimated.

Figure 1: Marker identification. The first loading is fixed at 1, and the intercepts are equated. PDF page 90.

model results for models of data that were explicitly generated to show latent mean differences (see Tables 4.1 and 4.2). As a reminder, the data-generated latent mean difference for Contrasts 1 through 3 were, on average, -0.40, -0.20, and 0 (i.e., a null difference), respectively. In the null model, the data-generated mean differences for Contrasts 1 through 3 were zero, on average, across all three contrasts. An average

Figure 2: The data-generating contrasts are −0.40, −0.20, and 0. PDF page 116.

The MIMIC line in `osf/analysis/mimic_model.R` has no free latent intercept, and lavaan then fixes that intercept at 0. The slopes on a large sample are about −0.15, 0.06, and 0.27. Adding the intercept returns −0.399, −0.200, and 0.000 on a sample of 100,000 per group (`simulation/mimic_large.R`, seed 7). The saved MIMIC summary is near −0.40, so the stored column is the free-intercept fit.

3.2 Six misses at 400 replicates, then three at 10,000

A 400-replicate screen of the ten $N = 240$ cells, multigroup and free-intercept MIMIC, true and null, put all 30 averages inside two combined standard errors and six rejection rates outside that band. The script is `simulation/validate_gaps.R`. The scores are `simulation/out/gap_scores.csv`. One of the six also missed Bonferroni and Holm: MIMIC, equal groups, null means, contrast 1, 2.26 against 5.18, $|z| = 3.77$. A binomial test of six misses in thirty trials at $p = 0.05$ has upper tail 0.0033 if the indicators are independent. The dependence caveat above applies to that figure as well.

The same ten cells at 10,000 replicates are `simulation/validate_10k.R`, with one fixed L'Ecuyer seed per cell in `simulation/out/seeds_10k.csv`. Five of the six short-run misses fall inside the uncorrected band, including the corrected miss, which comes out 5.01 against 5.18. Three rates remain outside:

Cell	Contrast	Printed	At 10,000	$\lvert z \rvert$
MIMIC, 45/5, true difference	2	23.37	22.11	2.12
MIMIC, 45/5, null	2	7.41	6.46	2.64
MIMIC, equal groups, null	3	5.32	4.53	2.59

Their averages match the page. Bonferroni and Holm leave none of the thirty rates outside.

```
# ---- Models of N = 240 ----
pacman::p_load(here, tidyverse, simsem, lavaan, purrr)
set.seed(1313)
# ---- Significance Model ----
## ---- Model Syntax ----
# 4 item model
pop_model_cond1 <- "
eta =~ NA*i1 + 0.7*i1 + 0.7*i2 + 0.7*i3 + 0.7*i4
eta ~~ .5*eta
i1 ~~ 0.51*i1
i1 ~ 1*
i2 ~~ 0.51*i2
i2 ~ 1*1
i3 ~~ 0.51*i3
```

Figure 3: Appendix A. The first intercept is written `1*`. The loading line contains both `NA*i1` and `0.7*i1`.

The scores are simulation/out/gap_scores_10k.csv.

3.3 The simsem 0.5.16 redraw, and the convergence rule

The dissertation names simsem 0.5.16. For a lavaan script, `sim()` in that version draws the rows with `lavaan::simulateData`, after setting the generator to L'Ecuyer-CMRG and the seed to the `exportData` seed. simulation/env_miss.R reruns the miss cells that way, at seed 13131, with lavaan 0.6.17. The summary is simulation/out/env_miss.csv.

On that stream the three rates are 22.53 against 23.37 ($|z| = 1.42$), 4.86 against 5.32 ($|z| = 1.49$), and 6.46 against 7.41 ($|z| = 2.64$). Two fall inside the uncorrected band. The third is the same 6.46 the hand-written generator produced. The lavaan converged flag and the finite-standard-error rule selected the same replicates in every cell of this redraw. No captured warning contained the word "gradient." Requiring nonnegative variances dropped replicates only in the multigroup equal-null cell, 9,774 of 9,927, and that subset does not move the rates onto the page. The BFGS refit in Section 5 counts 9,999 converged replicates in the null 45/5 cell and moves this rate from 6.46 to 6.51. It stays outside the band. Section 4 is that comparison after the refit.

4 What is still outstanding

This section answers which comparisons remain after the 10,000-replicate runs and the BFGS refit. One rejection rate does not match. Three smaller differences are listed after it, and they are not replication failures. The remaining comparisons match, and Bonferroni and Holm leave zero misses among the thirty rates.

The comparison that does not match is the MIMIC rejection rate for contrast 2, null means, group shares 45/45/5/5. The page prints 7.41. The simsem stream gives 6.51 from 9,999 converged replicates. The hand-written stream gives 6.47 from 10,000. The combined standard error is 0.36 percentage points on both streams. The two-standard-error band around 7.41 runs from 6.69 to 8.13. Both of our rates fall below 6.69. The standardized gaps are $|z| = 2.50$ (simsem) and $|z| = 2.62$ (hand). The two-sided p-values are 0.012 and 0.0088. Bonferroni on thirty tests uses 0.00167, and

Holm stops at the first ordered p-value, so this rate is inside both corrected bands. It is the only rate outside the uncorrected band. The other twenty-nine rejection-rate comparisons match. On the simsem stream that count includes the two rates the hand-written generator had left outside the band. The simsem redraw put both inside, and the BFGS refit leaves them inside.

Three differences remain, and none of them is a replication failure.

The multigroup average of contrast 3, equal groups, null means, is 0.0015 against the printed −0.0021. The printed standard error 0.0013 is the standard deviation of one replicate divided by $\sqrt{10{,}000}$, and $|z| = 1.98$. That gap sits on the boundary of the two-standard-error band and inside $z = 2.94$. The nonnegative-variance subset of the same cell is 0.0014, $|z| = 1.92$, inside the band. An earlier stream of the cell was 0.0023, $|z| = 2.40$. The simsem stream is the one on the boundary. It is not a replication failure.

Table 4.5, multigroup null, $N = 240$, condition 3, prints 9,675 and 95.75%. Dividing 9,675 by 10,000 gives 96.75. The rendered glyphs are 95.75, and the count is 9,675. Our count in the hand-written generator is 9,656 of 10,000, which is 96.56%. That rate is inside two combined standard errors of 96.75 and outside two combined standard errors of 95.75. The count and our simulation agree. The percent glyph is not the count divided by 10,000. That pair is not a replication failure of the convergence rate.

The page prints −0.4001 for the MIMIC average of contrast 1 at $N = 240$ in the equal true-difference cell. The public summary stores −0.399094. The significance count in that cell is 9,573 on the page and in the file. The fourth decimal differs, and the test count does not. That page-versus-file difference is not a replication failure.

The MIMIC convergence counts are not in this list. Under BFGS they match the printed 10,000 of 10,000 in seven cells, and the remaining cell is 9,999. Section 5 records that refit.

5 What we tried on the outstanding gaps

This section answers what was tried on the gaps, and what prompted each tool. R [1] and lavaan [2] are the fitting tools in the dissertation and in the OSF scripts. A structural-equation fit is a numerical optimization of a covariance model, which is work for R. The Lean proofs check the code identities and do not fit those matrices. simsem 0.5.16 [3] is the version the dissertation names, so the cells that had missed were redrawn in that version. OpenMx [5] was brought in as an independent structural-equation estimator, to test whether the lavaan `nlminb` shortfall was the optimizer or the data. BFGS was the alternate lavaan optimizer tried after that diagnosis. The convergence counts close under BFGS. The contrast-2 rejection rate does not. The public record still has no `.dat` files and no `sessionInfo` from the run that wrote the tables.

The dissertation names R 4.2.2, simsem 0.5.16, and lavaan 0.6.17, and `nlminb` with a maximum of 10,000 iterations. The operating system and `RNGkind` are absent from that description. The default since R 3.6.0, and the default here, is Mersenne-Twister, Inversion, and Rejection. `sim()` replaces that with L'Ecuyer-CMRG and then uses the `exportData` seed. simsem 0.5-16 was current from 2021-03-28 until 0.5-17 on 2025-04-02. lavaan 0.6-17 was published on 2023-12-20. The ten-cell Monte Carlo used R 4.3.3 [1]. R 4.2.2 was not installed. The L'Ecuyer generator is the one `sim()` sets in both of those R versions.

`exportData(program = "Mplus")` writes data files. The estimation scripts call `lavaan::sem`. The page quotes a lavaan warning and names `nlminb`. No Mplus output is on the OSF node. Had Mplus been the estimator, its default would fix the first loading at 1, and a default summary prints a convergence criterion of 0.00005 with at most 1,000 iterations. A negative residual variance is reported and the solution is still returned. The tables come from the lavaan scripts above. Mplus

is not installed. simulation/mgcfa_osf.inp mirrors the multigroup string and was not executed.

The OSF node q9yu8 was created on 2025-04-18. The files now on it were uploaded on 2025-09-20. They are the generation scripts, the lavaan analysis scripts, and the summary CSVs. There is no sessionInfo, no log, no .Rhistory, and no .dat replicate. The generation scripts call set.seed(1313) and pass seed 13131 to exportData. That call is commented out. One uncommented call, in the $N = 480$ condition-3 null script, names group sizes $(168, 168, 72, 72)$, the 35/15 split. The Monte Carlo here uses the printed shares.

GitHub, GitLab, and Zenodo searches, and a clone of the twenty public repositories under jjeffries13, are logged in simulation/out/code_search.md. Queries for the dissertation title, osf/analysis/mgcfa_model.R, and set.seed(1313) on that account returned no dissertation script. jjeffries13/APA_2024 reuses the loading idiom and seed 13131 for a ten- and fifteen-indicator missing-data study. Dynamic-Fit-Indices/DFI_Presentation.Rmd calls sessionInfo() and sinks DFIsessionInfo.txt. That file is not in the repository. The commits run from March through October 2022, the file is dated 9 March 2022, and the working directory is under /Users/jayjeffries/. Two teaching repositories last changed in February and June 2023 carry a badge for R 4.2.1. A May 2022 file has a commented Windows path. The paths those years read sit on that Mac home directory. No 2023 or 2024 repository contains a saved sessionInfo, an renv.lock, a packrat file, or a package DESCRIPTION.

The dissertation text remains the source for R 4.2.2. A Mac path in other projects leaves the simulation machine unnamed, and it supplies no lavaan or simsem version. A BLAS difference would move a converged estimate in the last digits. It would not move a rejection rate by a percentage point. The search did not produce the random stream. Under the named optimizer the MIMIC counts stay short of 10,000. The next subsection separates an optimizer setting from the data.

5.1 OpenMx on the unconverged MIMIC replicates

The nlminb counts were short of the printed 10,000. The question is whether those replicates are bad datasets or an optimizer setting. OpenMx is the independent estimator used for that question. BFGS is the lavaan optimizer tried because OpenMx left the same datasets unconverged. The script is simulation/openmx_check.R. It replays the hand stream from simulation/validate_10k.R and the simsem stream from simulation/env_miss.R, keeps every replicate lavaan did not converge, and keeps 25 converged replicates in each of the eight cells as a control. OpenMx 2.22.11 [5] then fits the same free-intercept marker model. The CRAN build's default optimizer is SLSQP. NPSOL is not compiled into it. CSOLNP was run as a second optimizer.

The indicator covariance of a failed replicate has a condition number between about 3.6 and 5.4, and its smallest eigenvalue is between about 0.34 and 0.45. The nlminb attempt did not return a negative variance, and it did not raise the gradient warning. Those replicates do not look like a singular sample or a Heywood case in the data.

Stream and cell	nlminb failed	SLSQP code 0	CSOLNP code 0	BFGS recovered
Hand, equal, true	12	0	1	12
Hand, equal, null	35	1	1	35
Hand, 45/5, true	14	0	0	14
Hand, 45/5, null	48	0	0	48
simsem, equal, true	12	0	0	12
simsem, equal, null	36	0	1	36
simsem, 45/5, true	20	0	0	20
simsem, 45/5, null	49	0	1	48

OpenMx status code 0 is a converged fit. Code 1 did not occur. The other failures returned code 6, the RED status: the first-order conditions were not met. On the control replicates the largest absolute gap between a lavaan contrast and the OpenMx SLSQP contrast is 2.2×10^{-5}.

The same failed datasets were refit in lavaan with BFGS, with nlminb allowed 50,000 iterations, and with lavaan's simple starting values. The longer run and the simple starts converged none of them. BFGS converged 225 of the 226. Seven cells then have 10,000 converged replicates. The simsem null cell with shares 45/45/5/5 has 9,999. Counting those BFGS solutions, the contrast-2 rate in that cell is 6.51 in the simsem stream and 6.47 in the hand stream, against the printed 7.41. The printed convergence count of 10,000 is reached by changing lavaan's optimizer. The printed rejection rate is not. The dissertation names nlminb, and under nlminb the count stays short of 10,000.

Jeffries and Bovaird (2026), DOI `10.1177/01466216261452255`, was not obtained as a PDF. Unpaywall, Europe PMC, and the Wayback Machine had no PDF on 2026-10-27. The OSF node is the analysis record for that title.

6 Software

This section answers which versions produced the fits and the proofs, and why each tool was used. The session files are `simulation/out/session_info.txt` and `simulation/out/env_session.txt`.

R is the language of the dissertation and of the OSF scripts, and each fit is a numerical optimization of a covariance model. That is why the replication runs in R rather than as a Lean proof. Lean checks the code identities in Section 2. lavaan is the package those scripts call, and the dissertation names lavaan 0.6.17, which is the version of every fit below. simsem 0.5.16 is the version the dissertation names. The miss-cell redraw installed that version. OpenMx was added because the lavaan `nlminb` runs fell short of 10,000 converged replicates, and an independent estimator can separate an optimizer failure from a bad sample. BFGS is a lavaan optimizer. It was tried after OpenMx returned those datasets unconverged, to see whether a different lavaan optimizer would accept them.

The Monte Carlo ran in R 4.3.3 (2024-02-29) [1]. R Core Team (2024) is the citation for that installation. The project page is r-project.org.

Every fit used lavaan 0.6.17 [2]. Rosseel (2012) is *Journal of Statistical Software* 48(2), pages 1–36, 10.18637/jss.v048.i02. The package page is CRAN. The development repository is yrosseel/lavaan.

The ten-cell run used simsem 0.5.17. The redraw used simsem 0.5-16 [3]. The `citation` entry for that 0.5-16 install is Pornprasertmanit, Miller, Schoemann, and Jorgensen (2021), R package version

0.5-16. The package page is CRAN. The tarball installed for the redraw is `simsem_0.5-16.tar.gz`. The development repository is simsem/simsem.

The convergence check used OpenMx 2.22.11 [5]. Neale and coauthors (2016) is *Psychometrika* 81(2), pages 535–549, 10.1007/s11336-014-9435-8. The package page is CRAN. The development repository is OpenMx/OpenMx. The script is `simulation/openmx_check.R`. The cell summary is `simulation/out/openmx_summary.csv`, and the recomputed rates are `simulation/out/openmx_rates.csv`. The CRAN source is that version. On this R 4.3.3 the file `omxData.cpp` calls `Rf_isDataFrame`, which this R headers file does not declare, so that one call was compiled as `Rf_isFrame`. The default optimizer in the build is SLSQP.

The code identities were checked with Lean 4.22.0 [4]. The toolchain file `lean-toolchain` reads `leanprover/lean4:v4.22.0`. de Moura and Ullrich (2021) is *Automated Deduction – CADE 28*, Lecture Notes in Computer Science, Springer, pages 625–635, 10.1007/978-3-030-79876-5_37. The release is v4.22.0.

Mathlib [6] is the tag v4.22.0. `lake-manifest.json` pins commit `79e94a09`, and that commit is the tag. The mathlib Community (2020) is the *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381, 10.1145/3372885.3373824.

7 Assumptions

This section answers which choices were supplied here rather than read off the page.

7.1 Random-number stream

The 400-replicate screen used L'Ecuyer-CMRG, `set.seed(13131)`, and one drawn seed per cell. The 10,000-replicate run sets a separate fixed seed in each cell. The simsem redraw uses seed 13131 inside `sim()`. These are not Jeffries's replicates.

7.2 Package versions

The ten-cell Monte Carlo used R 4.3.3 and simsem 0.5.17. The miss-cell redraw installed simsem 0.5-16 and kept lavaan 0.6.17. R 4.2.2 was not installed. Versions are in `simulation/out/session_info.txt` and `simulation/out/env_session.txt`.

7.3 Convergence

A replicate counts as converged when `lavInspect(fit, "converged")` is true and the three contrast standard errors are finite. That flag is the one the OSF script reads from the summary header. In the simsem 0.5-16 redraw the two rules selected the same replicates.

7.4 The first intercept

The Monte Carlo generates from `i1 ~ 1*1`, the token in the OSF script and in the null model on the page.

7.5 The MIMIC intercept

The MIMIC Monte Carlo frees the latent intercept. The posted line, which fixes it at 0, does not recover −0.40.

8 Ambiguities that were resolved

This section answers which readings were settled by a file, a fit, and a command.

8.1 The minus signs on the generating contrasts

The question was whether the text extract had dropped the minus signs on −0.40, −0.20, and 0. Figure 2 is the rendered page. The crop script is paper/figures/crop_pages.py, command `python3` paper/figures/crop_pages.py. The output crops are in paper/figures. The proof commit is `2dcc1c4`.

8.2 Which population produces −0.40

The question was whether the marker model on equal loadings 0.70 is the printed scale. The artifact is simulation/out/candidate_readings.csv, from `Rscript` simulation/candidate_readings.R `out`. The parsed string, with the first loading filled in as 1, matches. The equal-loading marker reading does not.

8.3 The public analysis script

The question was which identification produced Tables 4.1 and 4.3. The artifact is osf/analysis/mgcfa_model.R, retrieved 2026-10-27. The multigroup summary matches Table 4.1 to the printed digits. The Monte Carlo of that string is simulation/out/validate_10k.csv, from `Rscript` simulation/validate_10k.R `10000 out`, scored by `Rscript` simulation/score_gaps.R simulation/out/validate_10k.csv simulation/out/gap_scores_10k.csv.

9 Code and source availability

This section answers where each source was looked up, and what came back.

Place	Date	Result
Dissertation PDF, Appendix A	2026-10-27	Generation syntax. sha256 `a21df98c...fd469b3`
OSF node `q9yu8`	2026-10-27	Scripts and summaries. No `.dat` files, no `sessionInfo`
DOI `10.1177/01466216261452255`	2026-10-27	No PDF from the publisher, Unpaywall, Europe PMC, or Wayback
lavaan 0.6.17	2026-10-27	Used for every fit
simsem 0.5-16, CRAN archive	2026-10-28	Used in simulation/env_miss.R. Generator is `lavaan::simulateData`
GitHub, GitLab, Zenodo	2026-10-28	No dissertation script. Log in simulation/out/code_search.md
OpenMx 2.22.11, CRAN source	2026-10-28	Installed. One call in `omxData.cpp` compiled as `Rf_isFrame`. Script simulation/openmx_check.R. Summary simulation/out/openmx_summary.csv
MplusAutomation, Mplus	2026-10-27	Not installed

The dissertation file was retrieved from the Wayback `id_` object dated 2025-05-11, snapshot `20250511025154`. The help request filed before the OSF node was found is https://github.com/alectionary/vibe-math/issues/33. The fetch request for the 2026 article is https://github.com/alectionary/vibe-math/issues/34.

10 Next steps

The MIMIC convergence counts close under BFGS, as Section 5 records. The comparison that remains outside the uncorrected band is the one rejection rate, 6.51 and 6.47 against 7.41. Jeffries's `.dat` files, or a `sessionInfo` from the run that wrote Tables 4.1 through 4.5, are the check that would tell that rate apart from this stream. The three smaller differences in Section 4 are not that gap. An R 4.2.2 binary would test whether the L'Ecuyer draws differ from R 4.3.3. The 2026 PDF would show whether the journal version uses these tables.

11 Artifacts

- Code identities. File `LatentContrasts/Orthogonality.lean`. Command: `lake build`. Commit `2dcc1c4`.
- Moment fits. File `simulation/out/candidate_readings.csv`. Command: `Rscript simulation/candidate_readings.R out`.
- Large-sample MIMIC. File `simulation/out/mimic_large.csv`. Command: `Rscript simulation/mimic_large.R out`. Seed 7.
- Four-hundred-replicate screen. File `simulation/out/validate_gaps.csv`. Command: `Rscript simulation/validate_gaps.R 400 out`. Scores: `Rscript simulation/score_gaps.R simulation/out/validate_gaps.csv simulation/out/gap_scores.csv`.
- Ten-thousand-replicate run. File `simulation/out/validate_10k.csv`. Seeds: `simulation/out/seeds_10k.csv`. Command: `Rscript simulation/validate_10k.R 10000 out`. Scores: `Rscript simulation/score_gaps.R simulation/out/validate_10k.csv simulation/out/gap_scores_10k.csv`.
- OpenMx convergence check. File `simulation/out/openmx_summary.csv`. Command: `Rscript simulation/openmx_check.R out`.
- simsem 0.5-16 redraw. File `simulation/out/env_miss.csv`. Command: `Rscript simulation/env_miss.R 10000 out`.
- Code search. File `simulation/out/code_search.md`.
- Page crops. Directory `paper/figures`. Command: `python3 paper/figures/crop_pages.py`.

The dissertation PDF is local and gitignored. Its sha256 is `a21df98c20401491e6b4a1d00a21b6f7cee5551a00c04a89fb414ec6cfd469b3`.

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Table 4.5
Initial Model Convergence Rates by Condition & Model

N	Condition	MIMIC Frequency Converged (%)	MGCFA Frequency Converged (%)
Significance Models			
240	1	10000 (100%)	9886 (98.86%)
	2	10000 (100%)	9811 (98.11%)
	3	10000 (100%)	8929 (89.29%)
480	1	10000 (100%)	9989 (99.89%)
	2	10000 (100%)	9994 (99.4%)
	3	10000 (100%)	9805 (98.05%)
960	1	10000 (100%)	10000 (100%)
	2	10000 (100%)	10000 (100%)
	3	10000 (100%)	9986 (99.86%)
Null Models			
240	1	10000 (100%)	9932 (99.32%)
	2	10000 (100%)	9911 (99.11%)
	3	10000 (100%)	9675 (95.75%)
480	1	10000 (100%)	9998 (99.98%)
	2	10000 (100%)	10000 (100%)
	3	10000 (100%)	9999 (99.99%)
960	1	10000 (100%)	10000 (100%)
	2	10000 (100%)	10000 (100%)
	3	10000 (100%)	10000 (100%)

Figure 4: Table 4.5. At $N = 240$ and condition 3 the multigroup significance model converged in 8,929 of 10,000 replicates. The null block of the same table prints 9,675 and 95.75%.